



Verification of Ethics, Resilience & Integrity for Trusted AI

Open-Source LLM Red Teaming & Safety Certification Platform

python

3.12+

License

AGPL 3.0

OWASP

LLM Top 10 2025

EU AI Act

Compliant

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About VERITY

VERITY is an **open-source** Red Teaming & AI Safety platform designed to audit Large Language Models (LLMs) against adversarial attacks. It automates the entire safety certification lifecycle: from attack execution to judge evaluation and regulatory compliance reporting.



Features



Red Team Capabilities

- **Prompt Injection:** Instruction override and context manipulation attacks.
- **Jailbreaking (42+ Payloads):**
 - **Single-Turn:** FlipAttack (98% ASR), PSA (97% ASR), Policy Puppetry, TokenBreak, Time Bandit, Encoding, Framing.
 - **Multi-Turn:** Crescendo, Echo Chamber (90%+ ASR), Deceptive Delight, Skeleton Key, CCA, Bad Likert Judge.
- **System Leakage:** Detection of system prompt extraction vulnerabilities.
- **Providers:** Ollama, OpenAI, Anthropic, Google Gemini.



Blue Team Analysis

- **LLM-as-Judge:** Automated safety scoring (1-10 scale).
- **Bootstrap CI:** Statistical confidence intervals for Attack Success Rate.
- **Detailed Reasoning:** Explainable verdicts for each attack.



Compliance & Reporting

- **OWASP LLM Top 10 2025:** Full vulnerability mapping.
- **EU AI Act:** Articles 9, 14, 15 compliance checking.
- **Professional Reports:** Markdown, HTML, and JSON export.



Architecture Overview

CLI / API

Red Team Orchestrator			
Prompt Injection Agent	Jailbreak Single Turn Agent	Jailbreak Multi-Turn (Crescendo TAP/PAIR)	System Leakage Agent
LLM Providers (Ollama OpenAI Anthropic Google)			
LLM-as-Judge			
Compliance Module + Report Generator (OWASP 2025 EU AI Act)			

Quick Start

Installation

```
# Clone repository
git clone https://github.com/GiuseppeSPk/VERITY.git
cd VERITY

# Create virtual environment
python -m venv .venv
.venv\Scripts\activate # Windows
# source .venv/bin/activate # Linux/macOS

# Install dependencies
pip install -e .

# Configure environment
cp .env.example .env
# Edit .env with your API keys
```

Basic Usage

```
# List providers
verity providers list

# Run prompt injection attacks
verity attack --type injection --max 5

# Run full security audit
verity audit --target ollama --model llama3.1:8b --output ./reports
```

Testing SOTA Attacks

```
# Run the SOTA attack test suite
python test_sota_attacks.py --max-attacks 5

# Test specific technique
python test_sota_attacks.py --technique psa
python test_sota_attacks.py --technique echo_chamber
```

Security & Liability

This tool is designed for AUTHORIZED SECURITY TESTING ONLY.

-  **Authorized Use Only:** Only use VERITY against systems you own or have explicit written permission to test.
-  **Legal Compliance:** You are responsible for complying with all applicable laws and regulations.
-  **No Malicious Use:** Do not use this tool for unauthorized access, harassment, or illegal activities.

By using VERITY, you agree to use it ethically and legally.

Contributing

Contributions are welcome! Please see [CONTRIBUTING.md](#) for guidelines.

License

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Made with  for AI Safety Research.